

ANKUR RAJ

ENGINEERING LEADER | DATABASE SYSTEMS | PLATFORM & DEVELOPER INFRASTRUCTURE

Bengaluru, India | Open to international relocation

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PROFESSIONAL SUMMARY

Engineering leader with 13+ years building database systems, developer infrastructure, distributed services and production AI tools. Currently Director of Software Development at Oracle, leading four engineering teams across India and Mexico. Grew the organization from 4 to 18 people, developed engineering managers and technical leads, and delivered systems used across Oracle's database portfolio. Brings hands-on C/C++ and database-runtime depth to architecture, reliability, hiring and multi-team delivery.

LEADERSHIP & BUSINESS IMPACT

- Built an organization from **4 to 18** engineers, developing **four Engineering Managers** and **three Technical Leads**; now lead through **seven direct reports** across four teams.
- Scaled DBDev Assistant into a reusable engineering platform adopted by **40,000+** users and supporting **500+** workflows.
- Led a major compiler modernization program across Oracle Database, delivering approximately **\$3M** in recurring annual savings while managing compatibility and rollout risk.

PROFESSIONAL EXPERIENCE

Director, Software Development - Database Engineering Platforms

Oct 2025 - Present

Oracle Corporation | Bengaluru, India; teams in India and Mexico

- Own engineering strategy, hiring, roadmaps and delivery for platform teams serving Oracle Database, Exadata, GoldenGate, database security and cloud database services.
- Coordinate priorities with database development, compiler, cloud, security and release organizations; guide DBDev Assistant, Codex Skills, EvalBench and CodeRadar from engineering prototypes to operational platforms.
- Continue hands-on architectural reviews and technical stewardship of Oracle Database's KGH heap/memory-management subsystem.

Senior Software Development Manager

Oct 2020 - Oct 2025

Oracle Corporation | Bengaluru, India; teams in India and Mexico

- Scaled the engineering organization and established its manager/technical-lead structure while delivering code coverage, memory safety, compiler tooling and regression-selection programs.
- Expanded database-specific DKC memory checks and runtime diagnostics using AddressSanitizer, UndefinedBehaviorSanitizer and Valgrind; improved the ability to reproduce and investigate memory corruption.
- Led test-infrastructure work covering approximately **30,000** tests and **17,000** regression groups, bringing source coverage and failure history into regression decisions.

Software Development Manager

Sep 2018 - Oct 2020

Oracle Corporation | Bengaluru, India

- Managed Oracle Database Test Tools engineers; built shared foundations for coverage analysis, regression workflows and developer productivity.

Principal MTS / Senior MTS / Member of Technical Staff

Jan 2015 - Sep 2018

Oracle Corporation | Bengaluru, India

- Advanced through three individual-contributor levels; built Java source-coverage instrumentation using a JavaCC parser and abstract syntax tree, alongside C/C++ instrumentation and developer tools.

Software Engineer

Jul 2013 - Dec 2014

Samsung R&D Institute India | Bengaluru, India

- Developed production software in C and Java.

SELECTED ENGINEERING WORK

Database Memory Management (KGH) & DKC | Low-level C/C++ systems and reliability

- Technical owner of KGH, the heap/memory-management subsystem within Oracle Database. Led changes in performance-sensitive C/C++ runtime code and investigated allocator behavior, corruption and compatibility issues.
- Extended memory-safety workflows with database-specific DKC checks, sanitizer builds and repeatable diagnostics to find out-of-bounds access, stale pointers and heap corruption earlier.
- Led self-service memory-testing workflows, including memtool, so developers could reproduce and verify findings without repeatedly rebuilding the full database.

Code Coverage & Engineering Data Platform | Instrumentation, ingestion and analytics

- Architected source-level coverage across C/C++, Java, PL/SQL, Python and Perl, with custom LLVM/Clang and Java coverage instrumentation. The platform processes **multi-terabyte** execution data with Kafka-based ingestion, aggregation and reporting.
- Supported coverage analysis across 20K+ source files in a database codebase exceeding 3B lines, making test effectiveness visible to development and release teams.
- Contributed release-quality signals through coverage and regression intelligence; Oracle Database 23ai recorded approximately **80%** fewer customer-reported issues than the prior major release.

PickTest & LITESuite | Intelligent regression testing

- Developed PickTest to rank regression tests using code changes, coverage relationships and failure history, and LITESuite to construct smaller test suites; together they reduced unnecessary testing while preserving targeted risk coverage.

LLVM/Clang Compiler Modernization | Compiler toolchains and large-scale change

- Led technical investigation, compatibility fixes, staged migration and stakeholder coordination for Oracle Database's compiler modernization; realized approximately **\$3M** in recurring annual savings.
- Authored upstream LLVM/Clang PR #201079 proposing coverage counters for call continuation, addressing misleading coverage results after non-returning calls.

DBDev Assistant | Production AI platform for engineering

- Architected a distributed internal assistant using REST APIs, containerized services, Oracle Cloud Infrastructure generative AI and Oracle Database vector search; adopted by **40,000+** users with **500+** workflows built on the platform.
- Implemented automated knowledge ingestion, retrieval-augmented generation (RAG), hybrid keyword/vector search and source-grounded answers to help engineers investigate issues with traceable evidence.

Codex Skills Platform & EvalBench | Agent tooling and evaluation

- Built reusable coding-agent skills and tool orchestration for engineering workflows. EvalBench tests prompt, retrieval and agent changes with deterministic checks, model-based judgments and auditable evidence; extended evaluation into customer-facing use cases.

CodeRadar | AI-assisted security analysis for C/C++

- Built an agent-assisted source-security workflow combining issue discovery, adversarial checks, expert code review and sanitizer-backed validation before reporting findings.

TECHNICAL CAPABILITIES

Systems & languages C, C++, Java, Python, SQL/PLSQL, Linux, Oracle Database internals, performance and concurrency.

Platforms & infrastructure LLVM/Clang, GCC, JavaCC/AST, Kafka, REST APIs, Docker, CI/CD, ASan, UBSan, Valgrind.

AI & engineering leadership RAG, vector/hybrid search, coding agents, LLM evaluation; hiring, coaching, architecture, roadmaps.

EDUCATION & OPEN SOURCE

B.Tech, National Institute of Technology Karnataka (2009–2013) | github.com/llvm/llvm-project/pull/201079